

MAT499 / 599

Project 2: Finding your best Neural Network

March 4, 2026

DUE: 11:59pm on Friday March 20th, 2026

Submission: Please submit this project to canvas as an iPython Notebook (.ipynb file). The project is worth 25% and 20% of the overall grade for MAT499 and MAT599, respectively. While the submission is a python script, also focus on presenting the questions logically (write them out!), explaining what the code in each block does using comments, and answering the word questions fully in text blocks within the notebook.

Task: This project will attempt to model the sonar dataset from the UCI Machine Learning Repository (<https://archive.ics.uci.edu/dataset/151/connectionist+bench+sonar+mines+vs+rocks>). A csv file of this data is available on the GitHub for this course (https://raw.githubusercontent.com/benjaminmlucas/MAT499/refs/heads/main/module_2/sonar.csv). This dataset provides 60 features for each sound and these can be used to determine whether the sound was a rock or a mine.

Prior to running the following set a seed to 499 or 599 (your enrolled class) so that your results are the same for every run of your notebook. You will need to set the seed for the numpy.random, random, and torch libraries.

Question 1

We would like to fit a 2-hidden layer fully connected neural network to this data, using 128 neurons and 64 neurons in the hidden layers, respectively. Using 5-fold cross-validation, find the training loss, training error, validation loss, and validation error for this model architecture for each fold. You should train the model for 200 epochs with a mini-batch size of 16. Use the adam optimizer for this learning with the default learning rate.

Create a table that shows the metrics for each fold and their averages.

Comment on the average training loss/error and validation error/loss for this model.

Question 2

Now add dropout to the model for regularization with a rate of 0.3. Perform the same 5-fold cross-validation as above and create the same results table. Did dropout improve the model in your educated opinion? Why / why not?

Repeat this process with a weight decay of 0.001 (removing the dropout). Perform the same 5-fold cross-validation as above and create the same results table. Did weight decay improve the model in your educated opinion? Why / why not?

Did dropout or weight decay work better on this model? Why?

Question 3

Using the preferred model from question 2, experiment with the learning rate. Try at least 5 values and justify which is best for this problem and dataset.

Question 4

It is strongly recommended that you use cross-validation to justify all hyperparameter choices on this dataset. Why is this a better approach than a single train-validation split? And why is it specifically more important on this dataset?

Hint: Look at the Standard Deviation of the validation error across your 5 folds. What does this value tell you about the stability of your model's performance?